

LUIS PRATAMA

+61 420 767 008 | [LinkedIn](#) | [Portfolio](#) | ruishuluis@yahoo.com |

PROFILE

Graduate Mechatronics Engineer with experience in industrial automation, robotics, mechanical design and system integration. Currently working in automation support with exposure to warehouse automation and Autonomous Mobile Robot (AMR) systems. Experienced in SolidWorks, embedded systems, prototyping, testing and technical documentation, with an interest in developing further across mechatronics, automation and engineering design.

ENGINEERING EXPERIENCE

Automation Support Technician

July 2026 - Present

Priority Integrated Systems

- Support the operation and performance of warehouse automation and Autonomous Mobile Robot (AMR) systems across operational warehouse sites
- Diagnose and resolve faults across AMR and automation systems
- Support system recovery to minimise operational downtime
- Work alongside onsite teams to troubleshoot technical and operational issues
- Gain hands-on exposure to robotics, automation, sensors and integrated warehouse systems
- Support automation systems across customer sites as required
- Tools & Skills: Industrial Automation, Autonomous Mobile Robots (AMRs), Robotics, Troubleshooting, Fault Diagnosis, System Recovery, Sensors, System Integration, Technical Support

Engineering Intern

May 2024 - August 2024

Optik Consultancy / Byron Bay Technology

- Designed and created a functional nozzle housing for Byron Bay Technology, enabling multidirectional movement and auto shut off
- Modelled mechanism in SolidWorks with attention to tolerances, alignment, and precise fit
- Integrated servo motors and Arduino to automate nozzle control, reducing manual labour and improving process efficiency
- Collaborated with engineers on prototyping, testing, and mechanical-electrical system integration
- Gained hands-on experience in embedded systems, motion control, and design for manufacturing
- Tools & Skills: SolidWorks, Arduino, Servo Motors, Mechanical Design, Embedded Systems, DFM, Prototyping

CASUAL EXPERIENCE

Hospitality Staff

May 2022 - Present

Super Agent Agency

- Provide professional front-of-house support at high-profile hospitality events, ensuring smooth guest experiences and efficient service delivery
- Work collaboratively in dynamic team environments to coordinate event setup, food and beverage service, and breakdown
- Maintain high standards of presentation, communication, and time management across diverse event settings
- Adapt quickly to fast-paced and high-pressure situations, supporting events leads and client needs flexibility and reliability
- Developed strong interpersonal, multitasking, and customer service skills applicable across client-facing and team-based roles
- Skills: Communication, Teamwork, Time Management, Client Service, Adaptability, Event Coordination

Barista

May 2023 - Present

MilkMe Barista

- Prepare and serve high-quality coffee and espresso beverages in a fast-paced mobile café environment across various Sydney events
- Operate commercial espresso machines with consistency and attention to detail.
- Maintain cleanliness, stock levels, and setup of mobile coffee stations in line with food safety and hygiene standards
- Deliver friendly and efficient service while managing high customer volume during peak hours
- Skills: Customer Service, Espresso Preparation, Time Management, Food Safety, Teamwork

RELATED PROJECT

UR3e Bottle Sorting Cobot - Perception System

February 2025 - May 2025

Robotics Studio 2

- Built a ROS 2 node to capture and process RealSense camera images for color-based bottle-cap detection
- Created an OpenCV GUI with live sliders and click-to-sample tools for on-the-fly threshold tuning
- Designed a simple and adjustable interface to test color detection quickly and accurately
- Integrated to translating visual data into a format robot could understand for precise sorting
- Maintained real-time performance under varying lighting conditions
- Worked closely with team members responsible for robot motion and gripper to ensure everything worked together

SortBot Simulation - YOLO

February 2025 - May 2025

Artificial Intelligence in Robotics

- Built a PyBullet simulation of a uArm robot sorting colored boxes using AI-powered vision and motion planning
- Collected and labeled 1,000+ top-down images to train a real-time object detector, achieving over 95% box-identification accuracy in simulation
- Tested object detection and pick-and-place performance under different conditions
- Improved how the system responded by mapping camera and robot positions accurately
- Presented progress regularly to peers and incorporated feedback from supervisors

Camera Guidance Mechanism for Small-Diameter Conduits (NBN)

August 2025 - November 2025

Design in Mechanical and Mechatronics Systems

- Integrated a compact camera inspection system within a motorised conduit-traversal mechanism (Ø20–100 mm)
- Implemented Raspberry Pi–Arduino serial control for bidirectional motion using dual NEMA 17 stepper motors
- Developed a lightweight operator interface with live camera feed, motion commands, and image capture
- Validated system performance for continuous operation (≥ 2 hours)
- Supported prototyping, integration, and laboratory testing in simulated conduit environments

TECHNICAL SKILLS

Language	C/C++, Java, Python, Arduino
Systems	macOS, Linux (Ubuntu), Windows
Software	ROS 2, OpenCV, SolidWorks, AutoCAD, Fusion 360, MATLAB, Visual Studio Code

CORE SKILLS

Teamwork

- Collaborated with a multidisciplinary team of five engineers to develop and deliver a client-focused project.

Problem Solving

- Contributed to solution brainstorming for a sorting system during internship, with the proposed concept later developed into a prototype showcased in a client demo.

Communication

- Demonstrated strong communication skills by coordinating with both the client and team members, contributing to smooth workflow and successful project delivery.

Multitasking

- Managed multiple academic deadlines while maintaining part-time employment, demonstrating strong time management and personal responsibility.

Language

- English (IELTS - 7)
- Bahasa Indonesia (Native)

EDUCATION

Bachelor of Engineering in Mechatronics (Honours)
University of Technology Sydney (UTS)

February 2022 - December 2025

Diploma of Engineering
University of Technology Sydney College

March 2021 - December 2021

CERTIFICATION

- Responsible Service of Alcohol (RSA)
- Class C Driver's License

REFEREES

Available on request.